

## Assignment - SAT, LAT

Q12 State the spread of light in  
a) air b) glass

Ans a) In air : Approximately  $3 \times 10^8$  m/s or  
exactly 299,792,458 m/s  
b) In glass : Approximately  $2 \times 10^8$  m/s

Q1 Explain the following terms

i Incident ray, Reflected ray, Angle of incidence,  
Angle of reflection, Normal

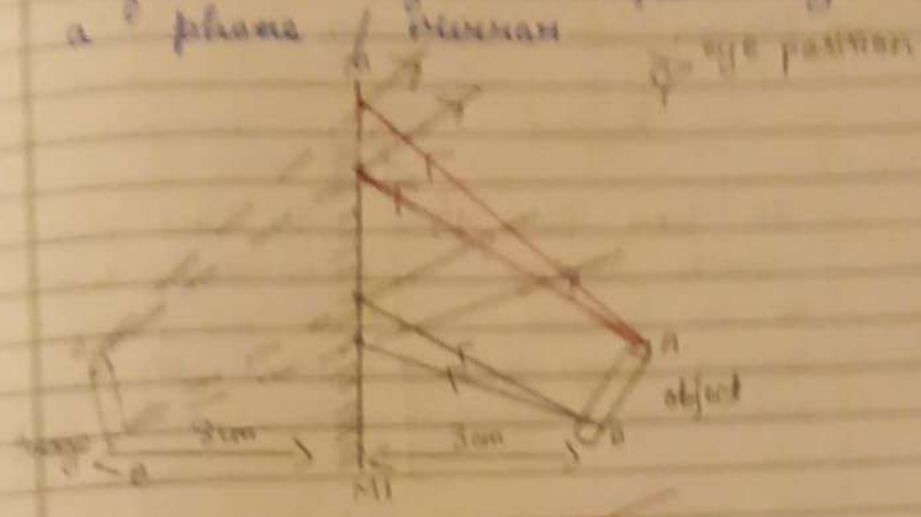
Incident ray - The ray of light which is  
returned back into the same, after reflected

Angle of incidence - Angle between the incident ray  
and the normal is called angle of incidence

Angle of reflection - The angle between the reflected  
ray and the normal is called the angle of reflection

Normal - The perpendicular drawn on the surface  
at the point of incidence, is called normal

Draw a ray diagram showing formation of an object by a plane mirror.



Draw the laterally inverted image of the following

- |      |      |
|------|------|
| a) L | a) J |
| b) B | b) B |
| c)   | c)   |

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Unit



Differentiate between the following:

i) Regular and irregular reflection.

| Regular                                                                                                                                                          | Irregular reflection                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>It takes place from smooth surfaces.</li> <li>All the parallel incident rays remain parallel after reflection.</li> </ul> | <ul style="list-style-type: none"> <li>It takes place from rough surfaces.</li> <li>All the parallel incident rays do not remain parallel after reflection.</li> </ul> |

Q. 8/10/16.

b) Real image and virtual image

| Real image                                                                                                                                                                   | Virtual image                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>It can be obtained on a screen.</li> <li>The rays of light after reflection meet at point.</li> <li>It is always inverted.</li> </ul> | <ul style="list-style-type: none"> <li>It cannot be obtained on a screen.</li> <li>The rays of light after reflection meet at a point.</li> </ul> |



It is formed in front of the mirror

• It is always erect.

• It is always found behind the mirror.

### Concept Map

Red Violet Indigo blue green yellow orange

↑ ↑ ↑ ↑ ↑ ↑  
Constituents of white light

Light

Rectilinear propagation

Reflection of light

Causes the formation of shadows

Types of Reflection

Law of reflection

Angle of incidence

Regular Reflection

Diffused or Irregular Reflection

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## Snap Test

1. A student directs a ray of light towards a plane mirror. They say makes an angle of  $40^\circ$  with the normal at the point of incidence.

a) What is an angle of incidence -  
Angle between the incident ray and the normal is called angle of incidence.

b) What is angle of reflection - Angle between the reflection ray and the normal is called angle of reflection.

c) State the two laws of reflection

- Angle of incidence is always equal to angle of reflection.
- Incidence ray, reflected ray and the normal all lie on the same plane.



2. A ray of light falls on a plane mirror along the normal.

a) what is the angle of incidence  
=  $90^\circ$  ✓

b) In which direction will the reflected ray travel.

= It will travel back at the incident ray. ✓

3. An incident ray makes a  $30^\circ$  angle with a mirror surface. Find the angle of reflection

= If incident ray =  $30^\circ$  reflected ray =  $30^\circ$  as  $\angle i = \angle r$  ✓

4. what is the speed of light

air =  $3 \times 10^8$  ✓

glass =  $2 \times 10^8$  ✓



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## ENERGY

### Keywords

1. Force
2. Conservation
3. Transformation
4. Hydroelectricity
5. Turbine
6. Armature
7. Generator
8. Dynamo
9. Magnetic Fields
10. newton
11. joule
12. Gravity
13. Calorie
14. Potential energy
15. Kinetic energy
16. Atomic energy
17. Nuclear energy
18. Electromagnet.



## i) Important Formula

work done = Force  $\times$  Distance moved  
in the direction of force.

## ii) Define the following terms:

- 1) 1 Joule - when a force 1 N displaces a body by 1m in the direction of force, the work is said to be 1 joule
- 2) Mechanical Energy - Energy due to the state of rest or motion of an object
- 3) Atomic or nuclear energy - Energy stored in an atom which is released due to the loss of mass when either a heavy nucleus splits into two lighter nuclei or combine to form a heavy nucleus.



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Subject : Biology



④  $d = 300 \text{ km}$

Average speed =  $60 \text{ km/h}$   
 $t = ?$

$t = \frac{\text{Total Distance}}{\text{Average speed}}$

$= \frac{300 \text{ km}}{60 \text{ km/h}}$

$= 5 \text{ km} \cdot \text{h}$